

Notes: Baqaee and Farhi (Econometrica, 2019)

The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem

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1 Big picture

- **Question.** How much do microeconomic productivity shocks matter for aggregate output once we move beyond Hulten’s first-order theorem?
- **Classical benchmark.** Hulten’s theorem says that, in an efficient economy, the first-order effect of a productivity shock to producer i equals its Domar weight λ_i :

$$d \log Y = \sum_i \lambda_i d \log A_i.$$

- **Core critique.** Hulten’s theorem is exact only locally. Since Domar weights and input-output linkages change endogenously after shocks, the first-order formula can miss important nonlinear macroeconomic effects.
- **Main contribution.** The paper derives second-order and nonlinear sufficient-statistic formulas for the macroeconomic impact of microeconomic shocks in efficient production-network economies.
- **Key mechanism.** Non-unitary elasticities of substitution make the economy asymmetric: complementarities amplify negative shocks and attenuate positive shocks, while substitutability does the opposite.
- **Network result.** Networks are not irrelevant once one moves beyond first order. The same Domar-weight distribution can imply different macroeconomic outcomes if the underlying production network and elasticities differ.
- **Quantitative result.** In a calibrated U.S. sectoral business-cycle model, nonlinearities create negative skewness, excess kurtosis, and nontrivial output losses even when sectoral shocks are symmetric and thin-tailed.
- **Applications.** The paper studies sectoral business-cycle shocks, the 1970s oil shocks, and long-run Baumol-style cost disease.
- **Bottom line.** Sales shares summarize first-order effects, but they are not enough for large shocks, tail risks, welfare costs, or long-run structural transformation.

2 Incremental contribution of the paper

- **Beyond Hulten’s theorem.** The paper keeps the generality and sufficient-statistic spirit of Hulten’s theorem but extends the analysis to second-order and global nonlinear effects.
- **Network macro with efficient equilibria.** Prior work often emphasized production networks in inefficient settings or used networks to explain covariance patterns. This paper shows that even in efficient economies, production-network details matter beyond first order.
- **General-equilibrium elasticities.** The paper introduces reduced-form general-equilibrium elasticities of substitution and maps them to structural microeconomic elasticities, input-output linkages, and factor reallocation.
- **Nonlinear amplification.** The paper explains how symmetric thin-tailed micro shocks can generate asymmetric and fat-tailed aggregate outcomes endogenously.
- **Disaster-risk contribution.** It provides a production-network mechanism for rare disasters without assuming fat-tailed exogenous shocks or fat-tailed Domar weights.
- **Quantitative discipline.** The paper takes the theory to U.S. input-output data, sectoral TFP shocks, oil shocks, and long-run TFP growth, showing that the nonlinear terms are quantitatively large.
- **Conceptual bridge.** It connects three literatures: growth accounting, production networks, and structural transformation. The same nonlinear logic explains short-run macro fluctuations, oil-shock amplification, and long-run Baumol cost disease.

3 Data, calibration, and measurement

3.1 Nature of the exercise

The paper is primarily theoretical. It derives general results for efficient economies and then illustrates the magnitude of the nonlinear terms using calibrated U.S. input-output data and sectoral TFP shocks. The empirical exercises are not reduced-form causal estimates; they are quantitative applications of the theoretical formulas.

Interpretation: The relevant “data” are model calibration inputs: Domar weights, input-output shares, elasticities of substitution, sectoral TFP shocks, and historical oil expenditure shares.

3.2 Sectoral input-output data

The business-cycle calibration uses 88-sector U.S. KLEMS annual input-output data from Jorgenson and collaborators, excluding government sectors. The data cover sectoral output and inputs from 1960 to 2005. The paper uses sector-level TFP series computed by Carvalho and Gabaix using Jorgenson-Gollop-Fraumeni methodology.

Interpretation: These data provide two central objects: the steady-state production network and the empirical distribution of sectoral productivity shocks.

3.3 Elasticity calibration

The quantitative model uses three structural elasticities:

- σ : elasticity of substitution across final consumption goods;
- θ : elasticity of substitution between value added and intermediates in production;
- ε : elasticity of substitution across intermediate inputs.

The benchmark calibration is

$$(\sigma, \theta, \varepsilon) = (0.9, 0.5, 0.001).$$

The main table also reports lower-elasticity, higher-elasticity, and near Cobb-Douglas cases.

Interpretation: Low elasticities make the economy more nonlinear because inputs are harder to substitute away from after adverse shocks.

3.4 Labor reallocation cases

The paper considers two polar labor-market cases:

- **Full reallocation:** a common labor factor can move costlessly across sectors.
- **No reallocation:** sector-specific labor cannot be reallocated across sectors.

The no-reallocation case is interpreted as a better short-run benchmark; the full-reallocation case is more appropriate for medium-run or long-run adjustment.

Interpretation: The reallocation margin is central because bottleneck sectors are more damaging when labor cannot quickly move to reinforce weak links.

3.5 Applications and measured shocks

The quantitative section studies three applications:

1. a calibrated multi-sector business-cycle model with stochastic sectoral TFP shocks;
2. the 1970s oil shocks, using oil's changing Domar weight and production decline;
3. long-run aggregate TFP growth from 1948 to 2014, comparing nonlinear, first-order, and second-order measures.

4 Model primitives and theoretical specifications

4.1 Hulten's theorem as the first-order benchmark

Let Y be aggregate output, A_i producer-level TFP, and λ_i the Domar weight of producer i . Hulten's theorem says

$$\frac{d \log Y}{d \log A_i} = \lambda_i. \quad (1)$$

Therefore, locally,

$$d \log Y = \sum_i \lambda_i d \log A_i. \quad (2)$$

Interpretation: Hulten's theorem is powerful because it does not require knowing the entire production network. At first order, sales shares are sufficient statistics.

4.2 Second-order logic

The paper's key object is the Hessian of aggregate output with respect to microeconomic productivity shocks:

$$\frac{d^2 \log Y}{d \log A_j d \log A_i}. \quad (3)$$

By differentiating Hulten's theorem, these second derivatives are derivatives of Domar weights:

$$\frac{d^2 \log Y}{d \log A_j d \log A_i} = \frac{d \lambda_i}{d \log A_j}. \quad (4)$$

Interpretation: If a shock changes sales shares, then first-order Hulten weights are no longer enough. The direction and size of Domar-weight responses determine nonlinear amplification or attenuation.

4.3 GE elasticities of substitution

The paper defines reduced-form general-equilibrium elasticities of substitution ρ_{ij} . These capture how relative Domar weights respond to productivity shocks. A schematic version is:

$$1 - \frac{1}{\rho_{ij}} = \frac{d \log \lambda_i}{d \log A_j} - \frac{d \log \lambda_j}{d \log A_j}. \quad (5)$$

Interpretation: These are not purely technological elasticities. They combine micro substitution, network linkages, demand, and factor reallocation in general equilibrium.

4.4 Idiosyncratic shock approximation

For a shock to producer i , a simplified second-order expansion is

$$\log Y \approx \log \bar{Y} + \lambda_i \log A_i + \frac{1}{2} \frac{d\lambda_i}{d \log A_i} (\log A_i)^2. \quad (6)$$

When goods are GE complements, negative shocks are amplified and positive shocks are attenuated relative to first order. When goods are GE substitutes, the opposite occurs.

Interpretation: This is the paper’s central intuition. Nonlinearities make the sign and size of shocks matter. A negative electricity shock is not simply the mirror image of a positive electricity shock.

4.5 Correlated shocks

For multiple simultaneous shocks, cross-partials matter:

$$\log Y \approx \log \bar{Y} + \sum_i \lambda_i \log A_i + \frac{1}{2} \sum_i \sum_j \frac{d^2 \log Y}{d \log A_j d \log A_i} \log A_i \log A_j. \quad (7)$$

Interpretation: The macro impact of two shocks is not generally the sum of their separate impacts. A shock to one sector can change the importance of another sector through the network.

4.6 Reduced form to structural primitives

The paper maps GE elasticities back to structural primitives in nested-CES production networks. The mapping uses:

- network shares Ω_{ij} ;
- Leontief-inverse objects Ψ_{ij} ;
- structural micro elasticities θ_j ;
- factor reallocation terms.

Interpretation: This is the theoretical bridge: reduced-form GE elasticities summarize macro propagation, but the paper also shows which microeconomic parameters generate them.

4.7 Network irrelevance benchmark

A special case restores network irrelevance. If productivity shocks are factor-augmenting and all micro elasticities are equal, then second-order effects depend only on Domar weights and the common elasticity. In a single-factor nested-CES economy, aggregate output has a closed form.

Interpretation: This result clarifies when Hulten-style aggregation remains powerful beyond first order. It also shows exactly which assumptions must fail for networks to matter nonlinearly.

5 Identification and theoretical strategy

5.1 Step 1: keep efficient equilibrium as benchmark

The paper works with efficient competitive economies. This means first-order effects are pinned down by Hulten's theorem. Any additional macro importance of networks must therefore come from nonlinear terms, not from first-order distortions.

5.2 Step 2: show sales shares are only local sufficient statistics

Since Domar weights respond to shocks, the first-order coefficient itself changes after a shock. The paper identifies the second derivative of output with the response of Domar weights.

5.3 Step 3: characterize the Hessian

The paper derives formulas for the Hessian of log output in terms of general-equilibrium elasticities and then maps those elasticities to microeconomic elasticities and input-output matrices.

5.4 Step 4: use special cases to separate mechanisms

Examples isolate two mechanisms:

- **Relative-share effect:** if goods are complements, negative shocks increase the shocked sector's Domar weight and amplify output loss.
- **Input-output multiplier effect:** if a bottleneck input becomes scarcer, the input-output multiplier can rise, further magnifying the shock.

5.5 Step 5: calibrate and compare to first-order benchmarks

The paper calibrates a U.S. sectoral model and compares the nonlinear solution to first-order and second-order approximations. It then applies the logic to oil shocks and long-run productivity growth.

Interpretation: This is not econometric identification. It is theoretical identification of nonlinear terms plus quantitative calibration to evaluate their magnitude.

6 Figures and table

6.1 Figures 1 and 2: Nonlinearities from relative shares and input-output multiplier effects

Read: Figure 1 plots aggregate output as a function of a producer's productivity in two reallocation cases. Figure 2 plots output when the input-output multiplier changes endogenously.

Interpretation: In Figure 1, with no labor reallocation and Leontief-like complementarities, negative shocks are strongly amplified while positive shocks can be muted. With full reallocation, the output function changes because labor can flow toward the bottleneck. In Figure 2, even a one-good roundabout economy becomes nonlinear because output uses itself as an input and the input-output multiplier changes with the shock.

Economic message: These figures explain why Hulten’s theorem can be misleading for large shocks. At first order, sectors with the same Domar weight look identical. Away from first order, the ability to substitute and reallocate determines whether losses become amplified.

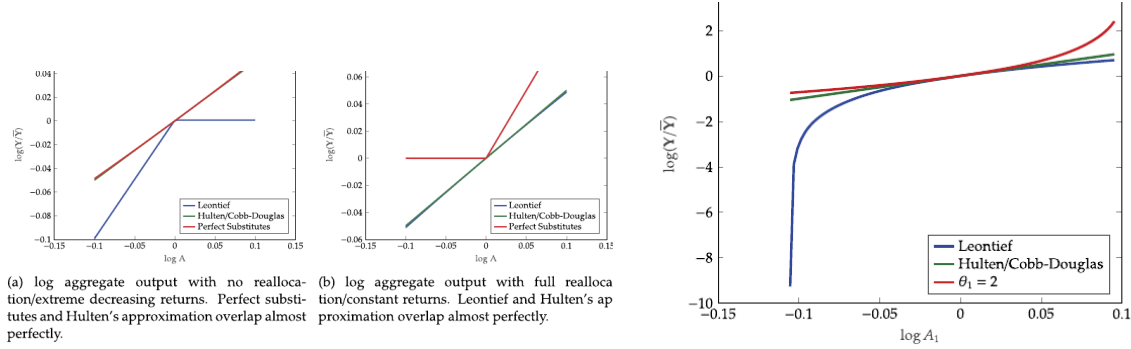


FIGURE 1.—log aggregate output as a function of productivity $\log A$, in the economy with full reallocation/constant returns for different values of θ_0 . This example consists of two equally sized industries using labor as their only input. The economies depicted in Figures 1a and 1b are all equivalent to a first order.

as a function of productivity shocks $\log A_1$ with variable input–output multiplier $\xi = 10$.

(a) Figure 1: log aggregate output with no/full reallocation.

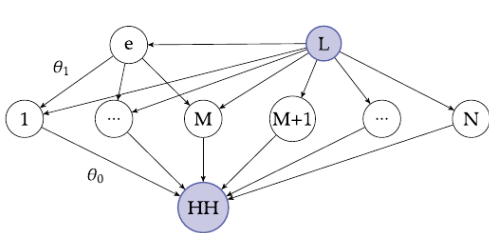
(b) Figure 2: output with variable input-output multiplier.

6.2 Figures 3 and 4: Energy examples with one factor and multiple factors

Read: Figure 3 shows an economy with energy as a near-universal produced intermediate input. Figure 4 shows an analogous economy where energy is treated as an endowment/factor.

Interpretation: Figure 3 emphasizes the upstream role of energy: downstream sectors need energy and labor, and household demand aggregates final goods. Figure 4 adds factor specificity: energy and labor are fixed supplies, so a negative energy shock cannot be offset by reallocating labor to the weak input. This makes energy a bottleneck.

Economic message: Oil and gas matter more than retail with the same Domar weight because energy is a near-universal bottleneck input. Its macro importance is not fully captured by sales share.



—An illustration of the economy with a near-universal intermediate input which can producer substitutes across labor and energy with elasticity $\theta_1 < 1$. The household final goods with elasticity of substitution $\theta_0 > \theta_1$. Energy is produced from labor

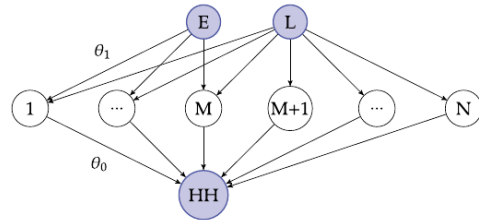


illustration of the economy with a near-universal input which we treat as energy. It has constant shares of labor and energy and substitutes across labor and energy with elasticity $\theta_0 > \theta_1$. Labor and energy are

(a) Figure 3: near-universal intermediate input.

(b) Figure 4: near-universal input as factor/endowment.

6.3 Table I: Simulated and estimated moments

Read: Table I reports mean, standard deviation, skewness, excess kurtosis, and volatility of Domar weights for full/no labor reallocation at annual and quadrennial horizons.

Interpretation: The no-reallocation benchmark with low elasticities generates negative means, negative skewness, and high excess kurtosis. The quadrennial no-reallocation case is especially nonlinear: it produces large left-tail risk because shocks are larger and labor cannot reallocate.

Economic message: Nonlinearities turn symmetric thin-tailed sectoral shocks into aggregate tail risk. This gives a production-network explanation for rare macroeconomic disasters without needing fat-tailed exogenous shocks.

SIMULATED AND ESTIMATED MOMENTS ^a					
$(\sigma, \theta, \epsilon)$	Mean	Std	Skew	Ex-Kurtosis	σ_A
Full Reallocation—Annual					
(0.7, 0.3, 0.001)	−0.0023	0.011	−0.10	0.1	0.090
(0.9, 0.5, 0.001)	−0.0022	0.011	−0.08	0.0	0.069
(0.9, 0.6, 0.2)	−0.0020	0.011	−0.05	0.0	0.056
(0.99, 0.99, 0.99)	−0.0013	0.011	0.01	0.0	0.001
No Reallocation—Annual					
(0.7, 0.3, 0.001)	−0.0045	0.012	−0.31	0.4	0.171
(0.9, 0.5, 0.001)	−0.0034	0.012	−0.18	0.1	0.115
(0.9, 0.6, 0.2)	−0.0024	0.011	−0.11	0.1	0.068
(0.99, 0.99, 0.99)	−0.0011	0.011	0.00	0.0	0.001
Full Reallocation—Quadrennial					
(0.7, 0.3, 0.001)	−0.0118	0.026	−0.4	0.4	0.307
(0.9, 0.5, 0.001)	−0.0113	0.026	−0.28	0.4	0.176
(0.9, 0.6, 0.2)	−0.0100	0.026	−0.23	0.2	0.133
(0.99, 0.99, 0.99)	−0.0058	0.025	0.01	0.0	0.003
No Reallocation—Quadrennial					
(0.7, 0.3, 0.001)	−0.0270	0.037	−2.18	12.7	0.404
(0.9, 0.5, 0.001)	−0.0187	0.030	−1.11	3.6	0.267
(0.9, 0.6, 0.2)	−0.0129	0.027	−0.44	0.7	0.154
(0.99, 0.99, 0.99)	−0.0057	0.025	0.00	0.0	0.002
Annual Data	—	0.015	—	—	0.13
Quadrennial Data	—	0.030	—	—	0.27

^aFor the data, we use the demeaned growth rate of aggregate TFP. For the model, we use the sample moments of log output. The simulated moments are calculated from 50,000 draws. Skewness and kurtosis of the data are impossible to estimate with enough precision and so we do not report them.

Table I: simulated and estimated moments.

6.4 Figures 5 and 6: Aggregate-output distributions and oil versus retail shocks

Read: Figure 5 compares the distribution of aggregate output in the benchmark nonlinear model with a log-linear/Cobb-Douglas model. Figure 6 compares shocks to oil and gas with shocks to retail trade.

Interpretation: Figure 5 shows that nonlinearities push mass into the left tail, especially at quadrennial frequency. Figure 6 shows that two industries with similar sales shares have different macro impacts: oil and gas generates much more damage for large negative shocks than retail trade.

Economic message: The shape of the network and elasticities determine which sectors are macro critical. Domar weights are insufficient for large shocks.

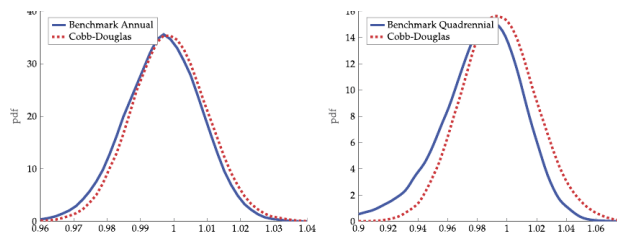
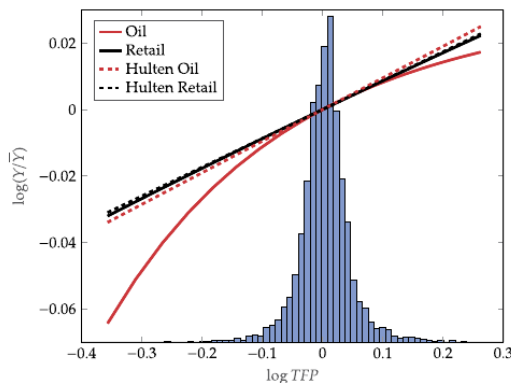


FIGURE 5.—The left panel shows the distribution of aggregate output for the benchmark model and linear model at an annual frequency. The right panel shows these for shocks at a quadrennial frequency. The difference between the two frequencies is that shocks are larger (their volatility is higher) at a quadrennial than at an annual frequency. The benchmark model has $(\sigma, \theta, \epsilon) = (0.9, 0.5, 0.001)$ and no labor reallocation. Note that the scales are different in these two figures.

(a) Figure 5: aggregate output distributions.



(b) Figure 6: oil and gas versus retail shocks.

6.5 Figures 7 and 8: Oil expenditure share and long-run TFP growth

Read: Figure 7 plots global crude-oil expenditure as a share of world GDP. Figure 8 compares nonlinear, first-order, and second-order measures of cumulative TFP growth from 1948 to 2014.

Interpretation: Figure 7 shows that oil's Domar weight rose dramatically during the 1970s oil shock, so averaging pre- and post-shock weights greatly increases the inferred impact relative to a first-order calculation. Figure 8 shows that first-order aggregation overstates long-run TFP growth relative to the nonlinear measure, while the second-order approximation tracks the nonlinear path much more closely.

Economic message: Nonlinearities are relevant both for short-run disasters and long-run growth. Baumol's cost disease is one manifestation: low-growth bottleneck sectors gain share and drag down aggregate TFP growth.

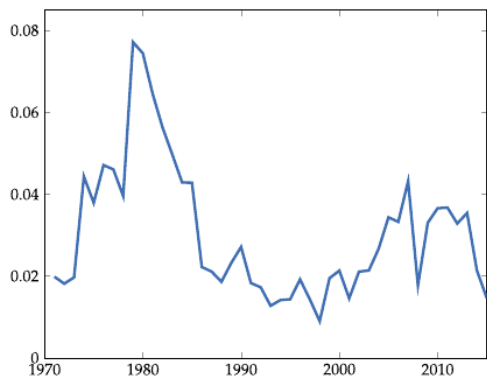


FIGURE 7.—Global expenditures on crude oil as a fraction of world GDP.

(a) Figure 7: crude-oil expenditure share.

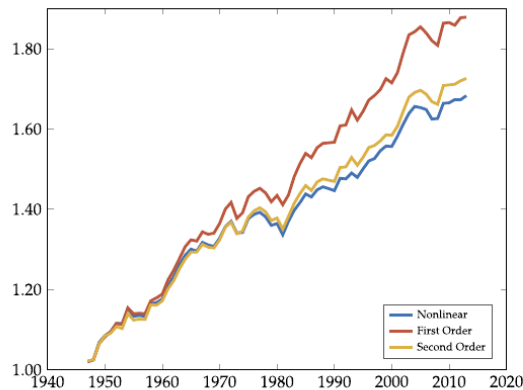


FIGURE 8.—Cumulative change in TFP: nonlinear (actual), first-order approximation, and second-order approximation.

(b) Figure 8: cumulative change in TFP.

7 Interpretive synthesis

- **First order:** Domar weights are sufficient statistics.

- **Second order:** changes in Domar weights are the new sufficient statistics.
- **Structural mapping:** changes in Domar weights depend on network structure, micro elasticities, and factor reallocation.
- **Macro implication:** bottleneck sectors matter disproportionately for large adverse shocks.
- **Tail-risk implication:** symmetric micro shocks can generate asymmetric aggregate outcomes.
- **Growth implication:** the economy’s changing sectoral composition can create nonlinear drag on aggregate TFP growth.

8 Short summary

- Hulten’s theorem gives the first-order impact of micro TFP shocks: the Domar weight.
- Baqaee and Farhi derive second-order and nonlinear formulas for efficient production-network economies.
- Non-unitary elasticities of substitution and input-output linkages determine how Domar weights change after shocks.
- Complementarities amplify negative shocks and attenuate positive shocks.
- Factor immobility increases nonlinear amplification by limiting reallocation toward bottleneck sectors.
- In U.S. sectoral calibrations, nonlinearities generate negative skewness and excess kurtosis in aggregate output.
- Oil shocks have outsized effects because oil is a bottleneck input whose Domar weight rises in crises.
- Long-run structural transformation produces a nonlinear drag on aggregate TFP growth, related to Baumol’s cost disease.

9 Conclusion

9.1 Core conclusion

Baqaee and Farhi show that Hulten’s theorem is a powerful but fragile first-order result. Once shocks are large enough to move sales shares, the details of production networks and elasticities matter for aggregate output, even in efficient economies.

9.2 Economic intuition

Sales shares tell us how important a sector is at the current equilibrium. But after a large shock, prices, quantities, and sales shares move. If the shocked sector is a bottleneck and substitution is hard, its sales share can rise after a negative shock, magnifying output losses. If factors cannot reallocate quickly, amplification is even stronger.

9.3 Incremental contribution in one sentence

The paper turns Hulten’s theorem from a first-order aggregation result into a nonlinear sufficient-statistic framework that explains tail risk, oil-shock amplification, and long-run productivity drag in production networks.

9.4 Final takeaway

Production networks are not just a story about who trades with whom. They are a nonlinear macroeconomic propagation mechanism. For small shocks, Domar weights may be enough. For large shocks, disasters, and long-run structural change, elasticities and network structure are essential.